

Real-time Animal Identification and Alert System using IoT and Deep Learning

Manujakshi B C

Department of Computer
Science and Engineering,
Faculty of Engineering and
Technology, JAIN (Deemed-
to-be-University), Bengaluru,
Karnataka, India

Shashidhar T M

Department of Computer
Science and Engineering,
Acharya Institute of
Technology, Bengaluru,
Karnataka, India

Ravikiran H N

Department of Electronics
and Communication
Engineering,
Adichunchanagiri University,
Nagamangala, Mandya,
Karnataka, India

Asha R

Department of Electronics
and Communication
Engineering, Vidya Vikas
Institute of Engineering and
Technology, Mysore,
Karnataka, India

Rajeev Bilagi

Department of Computer
Science and Engineering,
Acharya Institute of
Technology, Bengaluru,
Karnataka, India

Abstract The amount of traffic accidents that result in fatalities and injuries is one of the most urgent problems facing the world today. Many tragic incidents are caused by collisions between humans and animals. Worldwide, there are startlingly many automobile accidents—one every few seconds. Our primary objective is to decrease the probability of traffic accidents caused by animals by identifying them and informing cars of their presence. We suggest using IoT technology to create a detection and warning system that can significantly lower animal-related incidents. The purpose of this warning system is to alert cars on the roadway as soon as a detection event takes place by means of a signal. Our solution is based on Internet of Things technology, with the Raspberry Pi3 Model B acting as the main component for car warning and animal identification. Real-time photographs and videos of animal movements are captured by the Pi camera, which is equipped with a Raspberry Pi, through the use of sensors to identify impediments. After that, it alerts any vehicles and pedestrians using the woodland road. Additionally, it marks particular distances at night with lights. In the end, the document's primary goal is to safeguard numerous people from possible harm and to preserve animals from mishaps. In order to do this, the YOLO (You Only Look Once) algorithm was created. It processes approximately 45 frames per second and examines the complete picture in order to determine if objects are present or not.

Keywords: YOLO model, IoT, Deep Learning, Raspberry Pi

I. INTRODUCTION

The amount of road traffic accident-related deaths and injuries is a pressing global issue that we are currently facing. One of the primary causes of these regrettable incidents is collisions between humans and animals. Around the world, there are road accidents every fifteen minutes. Serious issues arise from human-animal conflicts, which frequently cause crop damage, human casualties, and human-killed animals. Specifically, died while driving through forests. Accidents still happen even if different

warning signs have been put up to lessen the incidence of animal-vehicle crashes. These collisions are caused by things like inattentive driving, a lack of awareness, and a failure to notice signs. The development of animal detection applications is essential to the resolution of practical issues. The majority of these apps rely on the ability to recognize animals in pictures or videos. Thus, our goals are to save animals, lower the global death toll, and lessen the number of accidents-related wild animal deaths. Reducing animal-vehicle collisions is the primary goal of animal detection. Our goal is to create an accurate and efficient system that may be deployed inside cars or at regular intervals along highways to alert drivers to the presence of animals and request that they slow down

A. Internet of Things

A network of interconnected computing devices, digital and mechanical machines, objects, animals, or people is referred to as the Internet of Things (IoT). Each of these devices is given a unique identity and is capable of transmitting data over the network without the need for human intervention, such as in human-computer interaction. Internet of Things-based technologies have the potential to revolutionize daily life and offer enhanced services in the near future.

Technological advancements in medical, energy, agriculture, gene therapy, smart cities, and smart homes are a few areas where they are strongly present. Currently, the network is home to approximately 9 billion items. This number is predicted to increase to as much as 20 billion with the advent of the Internet.

A. IoT relies on four principal components

- Low-power embedded systems: When designing electronic systems, these systems put great performance and low battery usage first.
- Cloud computing: Reliable storage servers are necessary due

to the high volume of data created by Internet of Things devices. Processing and analyzing the data to generate insights—such as locating electrical faults or system errors—is made possible in part by cloud computing.

iii. Large data availability: Since real-time sensors are essential to the Internet of Things, their widespread use in a wide range of applications will inevitably result in a rise in large data.

iv. Network connectivity: Every physical thing requires an IP address-based Internet connection in order to communicate effectively. To represent real objects, researchers must investigate alternate naming systems because the existing system runs out of IP addresses as the number of devices rises.

B. Deep Learning

Within the study of artificial intelligence and machine learning, deep learning is a subset of machine learning. It deals with networks that are capable of autonomous learning from unlabeled or unstructured data. The functioning of deep neural networks, deep learning, and deep neural learning is strikingly similar to that of the human brain. Deep learning algorithms are characterized by their capacity to continually examine structured data and deduce human thinking by using a logical framework. Deep learning makes use of a multilayer algorithm known as a neural network to carry out this purpose.

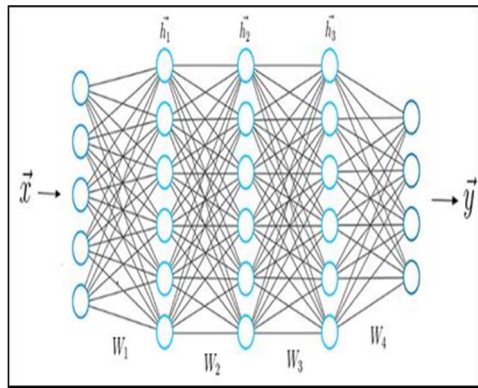


Fig. 1: Neural Network

A neural network's fundamental architecture, comprising the input and output layers, is depicted in Fig. 1. A neural network contains one input and one corresponding output for each input layer.

The fundamental building blocks of a neural network are the neurons within it. These networks are made up of hidden neurons in the network's hidden layer, input neurons that correspond to the input layer, and output neurons that correspond to the output layer. To generate the appropriate output, the hidden neurons apply weights to the input and then send it via an activation function. They route the data after applying a nonlinear change to it.

II. RELATED WORK

The research work mentioned in [1] includes the main
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features and results of the DETER-R (Deforestation Protection Near Real-Time) system, an automated deforestation detection system centered on the Brazilian Amazon, based on Synthetic Aperture Radar (SAR) are presented in this article. [2] Deforestation and natural disasters (e.g. forest fires, increased gas emissions) pose several serious challenges to forest ecosystems. This study proposes the use of analog and digital sensors, signal analysis techniques, and Raspberry Pi Model 3 as the basis for an intelligent forest environmental monitoring system. In addition to their great potential to slow down human-induced climate change, forests [3] provide many other benefits to society. However, in the 21st century, 98 forest carbon sinks could be seriously threatened by climate change-related challenges. Here we summarize current knowledge on climate-driven threats to forest stability from fire, drought, biological agents, and other disturbances⁹⁹. Recent developments, including in [4], are near-real-time (NRT) forest monitoring and warning systems that use remote sensing data to support the management and protection of forest resources. Local response teams tasked with assessing and mitigating threats to tropical forest ecosystems can benefit greatly from rapid and accurate detection of deforestation-related activities using satellite and airborne remote sensing. This study [5] proposes a versatile, easily deployable, and scalable wildfire detection and management system based on sensor networks (SNs) that can be used in a variety of environments and regions. SNs are deployed in hazardous and sensitive locations to monitor the environment and provide relevant data for analysis. Actively detecting and managing wildfires can be challenging in certain situations, especially in isolated and difficult-to-access areas or within fire hotspots.

III. METHODOLOGY

The suggested system architecture design is displayed in Figure 2. Our system's accuracy is mostly dependent on the ultrasonic sensor's capacity to recognize objects in its immediate surroundings, including people and animals. Upon identifying an animal, the sensor triggers the camera to record video. After being encoded into frames, the captured video is supplied to the model. YOLO is an efficient and accurate image detection algorithm that is used to train the model on the animal dataset. After identifying an animal in the picture, the model classifies the animal and outputs its position, which is subsequently entered into the database using the Flask backend.

In addition, a signal is transmitted to turn on the buzzer and LED lights as a warning system when an object is detected. The driver warning application's backend continuously searches the database for further new asset items before sending the information to the system's frontend, which plots it on a map. In order to prevent a potential accident with the wild animal, the driver is alerted by an auditory warning and receives a map showing the animal's location. This specific technique has the ability to greatly lower the frequency of collisions between humans and animals, hence reducing the number of traffic accidents brought on by wild animal interactions.

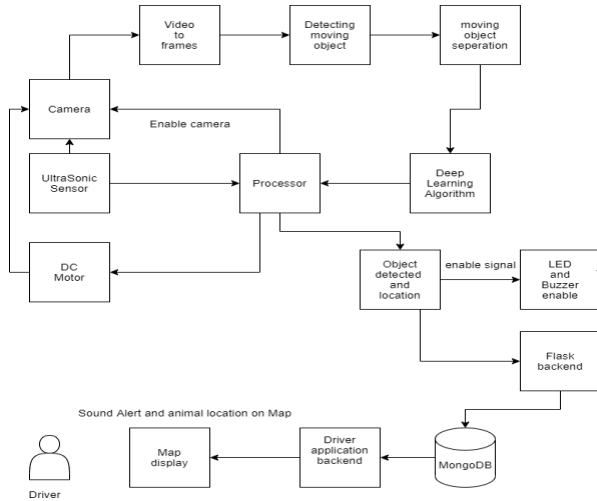


Fig. 2: Proposed System Design

IV. RESULTS

In this study, we employ an ultrasonic sensor and a Raspberry Pi device with a Pi camera to detect animals, as illustrated in Figures 3, 4, and 5. The major goal is to identify wildlife and alert approaching motorists to reduce the number of wildlife-related accidents on forest routes. The primary warning systems for cars include a driver alert app, LED lights, and a buzzer. DC motors are also used by it for particular rotational tasks. Every component of the system is thoughtfully positioned at key points along the woodland trail. An ultrasonic sensor recognizes the presence of an animal as it gets closer to the road and triggers the Pi camera. In order to classify the animal, the camera then captures a live image of it and feeds it to a deep learning model. Both an LED light and an audio buzzer sound when the sensor triggers an alarm, warning other cars in the area. Data about the locations and classification of animals is kept in a database. An alert program keeps a constant eye on this data, checking every second for new entries. The animal's location is indicated on the map, and an audible alert is produced by the map display. In the end, this method lowers the number of accidents on forest routes and the harm done to both wildlife and cars by encouraging cars to slow down and give animals enough space to cross or move across the road.



Fig. 3: Animal class Elephant Object Detection



Fig. 4: Animal class Tiger Object Detection

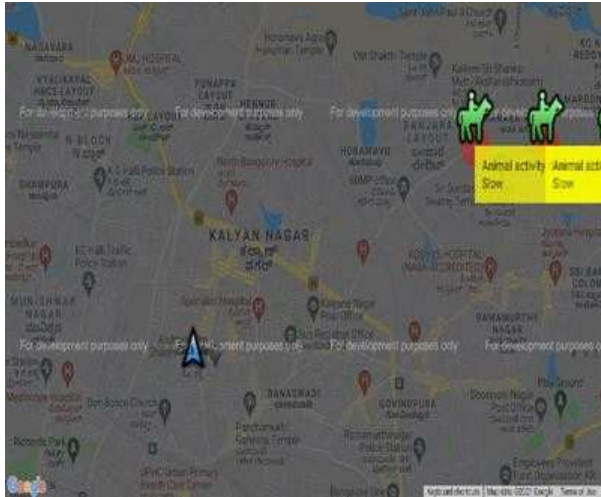


Fig. 5: A Driver Alert Application Map

V.CONCLUSION

Machine learning techniques, particularly deep learning, have become indispensable in many facets of our life in today's technologically advanced society. Numerous algorithms and applications employ these techniques. This work combines deep learning with IoT to propose a novel method for detecting wildlife. The primary objective is to recognize wildlife and promptly alert passing cars on forest roads. It is well known that forest road accidents frequently result in injury or death to animals and negatively affect endangered species. The Raspberry Pi3 Model B is used in our project to identify items, such as animals, and alert passing cars to their presence. We accomplish this by detecting obstructions using an ultrasonic sensor. The Pi camera detects obstacles and turns on to capture real-time images of the animals. Deep learning techniques are then used to examine these photos; these methods are kept in a database. The car on the forest road is then alerted to possible danger. Future developments might see the system integrated into automobiles, its detecting range expanded, and its autonomous speed lowered to prevent crashes. Our goal is for our study to help prevent accidents and conserve animals.

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